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High-precision fluidic Kirigami morphing surface for ultrasonic holographic lensing and haptic interfacing

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ABSTRACT

Morphing surfaces provide a versatile tool to advance the functionalities of high-performance aircraft, soft robots, biomedical devices, and human–machine interfaces. However, achieving *precise* shape transformation and mechanical property control remains challenging due to nonlinearity, design constraints, and the difficulty of coordinating multiple constituent materials across a continuous surface. To this end, this study unveils that Kirigami art can inspire novel solutions. More specifically, the geometric principles of Kirigami can be exploited to design and fabricate fluidic morphing surfaces capable of highly accurate shape morphing and output force control, all via a single pressure input. This study presents a systematic approach to designing Kirigami for tuning the local deformation and force output through extensive nonlinear modeling and experiment validation on two archetypal patterns: concentric square and circular cuts. The potentials of this approach are demonstrated via two multiphysics case studies: (1) an acoustic holography lens for ultrasonic wave steering, achieving highly accurate deformation control under a single global pressure input, and (2) a haptic device with a small volume, constant contact area, and high-resolution output force. The fluidic Kirigami concept allows for simple yet effective adaptation to different shape and stiffness requirements, paving the way for a new family of scalable and programmable morphing surfaces.

1. Introduction

Soft and morphing surfaces have shown exciting capabilities to create and augment the interfaces between widely different physical domains [1,2] — between fluid and solids [3–5], human and machines [6,7], or bodies and their environments [8–10]. By transforming its external shape or internal architecture, these morphing interfaces can be used to engineer critical interface behaviors like wave propagation [11–15], electromagnetic wave manipulation [16–18], haptic feedback [19,20], boundary flow development [21–23], and object manipulation [24–28]. As a result, morphing surfaces are among the key drivers behind recent advances in soft robotics, adaptive acoustics, aerospace, and biomedical industries.

Among these morphing concepts, the pneumatically inflatable morphing surfaces have garnered significant interest due to their simplicity, versatility, and safety [29–31]. These morphing structures typically feature carefully designed architecture with internal cavities, so that the pressure-induced volume expansion of these cavities can drive the overall shape change. The geometrical design and material selection of the structures enclosing the expanding cavities can directly dictate the final shape, providing considerable design flexibility. However, the current focus of inflatable morphing surfaces has primarily been on achieving accurate shape changes and their related applications.

In this study, we aim to develop an inflatable morphing surface that allows for precise control over *both shape morphing and force (or stiffness) output*. Additionally, we will explore new interdisciplinary and multi-physics applications. To this end, we present a concept by harnessing the out-of-plane deformation mechanics of Kirigami ("cut-paper" in Japanese). The key idea here is to wrap an automatically designed, simple-to-fabricate Kirigami sheet outside a soft-capped rigid fluid-filled container, creating a "fluidic Kirigami morphing surface" that can deform into a pre-programmed 3D topology under a global pressure input (Fig. 1a,b). Kirigami cutting is a reliable and scalable fabrication approach. It can be implemented on a wide selection of thin-sheet precursors of different sizes, including graphene [32–34], electronics [35–38], adhesives [39–41], and metals [42,43]. Cutting

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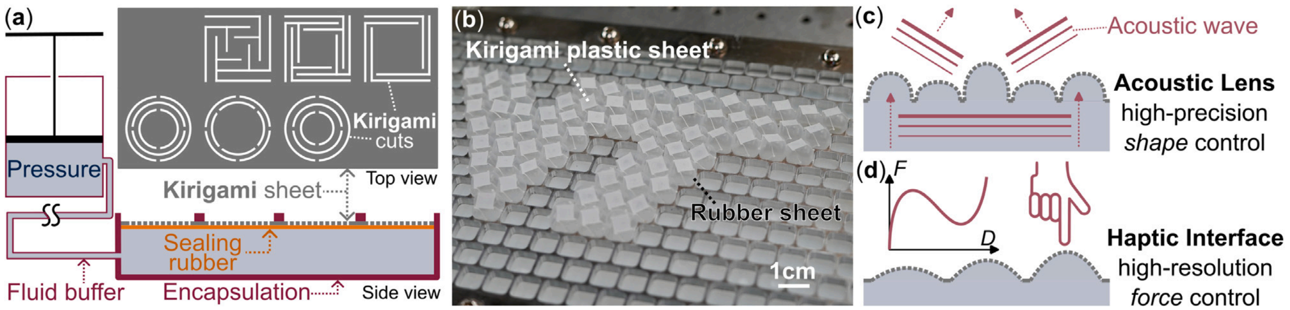

Fig. 1. The overall concept of fluidic Kirigami morphing surface. (a) By wrapping Kirigami cut sheets outside, and pressurizing a fluid-fluid container, we can create a morphing surface. Any thin-sheet material that can be easily cut may be used for Kirigami, and the cutting can be done with off-the-shelf machines, such as a plotter cutter or a laser cutter. (b) Our proof-of-concept tests showed that fluidic pressure can deform the Kirigami into a 3D topology (Video S1, Supplementary Materials), and the design of the Kirigami dictates such deformation. Therefore, we can leverage the correlation between design and mechanics behind the Kirigami cuts to achieve high-precision shape control (e.g., for acoustic focusing shown in (c)) or high-resolution force control (e.g., for haptic interface shown in (d)).

can be achieved with very high precision and almost limitless design possibilities [44–46]. More importantly, Kirigami's mechanical behavior — such as its external deformation under load, auxetics, and stiffness — is directly dictated by the underpinning cut pattern design [47–49]. Therefore, the nonlinear mechanics of the Kirigami sheet offer a reliable mechanism to precisely control the shape and elastic properties of the morphing surface.

This study demonstrates the high-precision outputs of fluidic Kirigami morphing surface via two inter-disciplinary functions. *The first function is acoustic holographic lensing, which presents demanding shape control requirements* (Fig. 1c). By carefully designing the Kirigami cut pattern and its corresponding 3D deformation under pressure, the fluidic Kirigami morphing surface can transform into an acoustic holographic lens, focusing ultrasonic waves in a targeted direction. If a new acoustic focusing pattern is needed, one only needs to peel off the Kirigami sheet and attach a new one with a different cut pattern. Accurate manipulation of acoustic waves, especially ultrasonic ones for medical applications, requires precise deformation of the morphing surface within the wavelength (typically in the millimeter range) [50,51]. For this reason, acoustic lensing is an ideal case for validating the high-precision shape control on the morphing surface using the Kirigami principle. It is also worth noting that in this demonstration, the fluidic Kirigami can also be categorized as a *metasurface* because it leverages its unit cells' thicknesses, which are at the sub-wavelength scale, to modulate the propagating wave's phase shift.

The second function is the haptic interface, which presents a demanding high-resolution force control requirement (Fig. 1d). Precise, adaptable, and customizable haptic feedback devices require force–deformation relationships tailored to desired human muscular and sensory preferences while situated in potentially small areas (e.g., below one's fingers or toes). This precise force feedback for targeted displacements is crucial in applications such as virtual reality, robotic surgery, and human touch-related neurophysiological investigations [52–55]. For each of these applications, high-resolution force output may be needed with minimal displacement to convey realistic physical interactions to a user. The inclusion of fluidic Kirigami enables such interactions, given its ability to deform out-of-plane in response to fluidic changes, such as air, with a mathematically modeled stiffness. An additional benefit of haptic devices created using fluidic Kirigami is that the design can be relatively small in size, with the physical interaction arising due to small inflation, followed by deflation of an object (e.g., rubber membrane of a small balloon/bubble).

To the best of our knowledge, this is the first time Kirigami is adopted in acoustic holographic lensing and highly precise application of force in haptics, so the results of this paper broadens the appeal of Kirigami engineering in general. More critically, achieving these two widely different functions with high precision requires mechanics modeling that can accurately correlate the Kirigami cut pattern design with its deformation and force output. Critically, this analytical model is the prerequisite for automatic and efficient design. Therefore, the following part of this paper will start by detailing the new nonlinear mechanics model of the Kirigami cell, then explain the inverse design process and demonstrate its use in acoustic lensing and haptic interfacing.

In particular, we will use a square-cut pattern for the acoustic lensing, and circular-cut for haptic feedback (Fig. 1). The square Kirigami pattern cuts the thin sheet material into a grid of unit cells consisting of narrow and straight ligaments, each deforming out-of-plane like a bent cantilever beam. While the circular-cut Kirigami's unit cell has curved ligaments that also bend out of plane. Moreover, the square and circular-cut Kirigami differ more than the shape of their constituent ligaments; the topology of interconnection between these ligaments is also fundamentally different. In the circular-cut Kirigami, each curved beam connects to other beams in the same ring via their ends, and it also connects to beams in the adjacent rings at its center. In other words, each ring in the circular-cut Kirigami is continuous without any breaks. In the square-cut Kirigami, on the other hand, each straight beam only connects to adjacent beams at the inner and outer layers. Because of the lower degree of connections between the straight ligaments, the square-cut pattern has a larger displacement range, making it more suitable for displacement control as required in the first application — acoustic holography. In contrast, the circular pattern offers relatively limited displacement range but greater load capacity, making it a better fit for high-resolution force control in the second application, the haptic interface.

Despite the differences between these two Kirigami cut patterns, we show that their nonlinear mechanics behaviors are governed by a few universal parameters that correspond to the geometry of the smallest ligaments, specifically, the beam width, length, and thickness, along with the number of beams. These parameters can be mapped into design spaces for practical implementations.

2. Square-cut Kirigami surface for high-precision shape control

The square-cut Kirigami cell consists of concentric "rings" of straight and interconnected thin ligaments of length l and width w (Fig. 2a). Therefore, a square-cut Kirigami cell can be fully defined by the following geometric parameters: the unit cell size d, the number of rings N, the ligament width w, and the gap between the cuts g. Notice the length l of each ligament can be directly calculated based on these parameters.

2.1. Analytical tool for designing the square-cut Kirigami cells' deformation

In this study, we focus on Kirigami cells with only one or two rings of ligaments ($N = 1, 2$) and model each ligament as a straight cantilever beam with compound loading at the free end (Fig. 2b,c). The *Euler–Bernoulli beam theory* describes their finite and geometrically non-linear deflection by:

$$M(x) = EI\frac{d\theta}{ds} = EI\frac{\frac{d^2y}{dx^2}}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}, \tag{1}$$

where $M(x)$ is the reaction bending moment along the beam span, θ is the beam cross-sections' rotation angle, s is the position along the deformed beam arc, x is the position along the initial, non-deflected beam, and y is the out-of-plane beam deflection. Assuming the nonlinear cantilever beam is subject to an external, transverse force P_0 and bending moment M_0 at its tip, Eq. (1) can be re-formulated into [56]:

$$\begin{aligned} \alpha &= \pm\frac{1}{\sqrt{2}}\int_0^{\theta_0}\frac{d\theta}{\sqrt{\lambda-\sin\theta}},\\ \beta_x &= \frac{x_0}{l} = \pm\frac{1}{\alpha\sqrt{2}}\int_0^{\theta_0}\frac{\cos\theta\, d\theta}{\sqrt{\lambda-\sin\theta}},\\ \beta_y &= \frac{y_0}{l} = \pm\frac{1}{\alpha\sqrt{2}}\int_0^{\theta_0}\frac{\sin\theta\, d\theta}{\sqrt{\lambda-\sin\theta}}, \end{aligned} \tag{2}$$

where α is a non-dimensional force variable. x_0 and y_0 are the horizontal and vertical beam displacements at its tip, respectively, so β_x and β_y are normalized tip deflections. θ_0 is the beam's rotation at the tip (Fig. 2c). The $\pm$ sign is positive (or negative) if the beam slope monotonically increases (or decreases). λ is a non-dimensional angle variable in that:

$$\begin{aligned} \lambda &= \sin\theta_0 + \kappa,\\ \kappa &= \frac{1}{2}\frac{M_0^2}{P_0 EI}. \end{aligned} \tag{3}$$

The generic solution of Eq. (2) when $0 \le |\lambda| \le 1$ is provided in Section 1 of the Supplementary Materials. By closely examining the deformation pattern of the square-cut Kirigami cell (Fig. 2b), we can assume that P_0 is positive, M_0 is negative, and $\theta_0 \approx 0$ (therefore, $\lambda \approx \kappa$ in Eq. (3)). Notice that if the external force is sufficiently high, θ_0 will deviate away from 0. However, this $\theta_0 \approx 0$ assumption can provide us with accurate predictions without incurring unnecessary complex formulations. Based on these assumptions, we can apply elliptic integral solutions to solve Eq. (2) so that:

$$\begin{aligned} \alpha &= f^*,\\ \beta_x &= \frac{x_0}{l} = \frac{2\sqrt{2\kappa}}{\alpha},\\ \beta_y &= \frac{y_0}{l} = \frac{f^* - 2e^* + 2\sqrt{2\kappa}}{\alpha}, \end{aligned} \tag{4}$$

where f^*, e^* are the complete elliptic integral of the first and second kind, respectively:

$$\begin{aligned} f^* &= 2F(\gamma,k) \equiv 2\int_0^{\gamma}\frac{1}{\sqrt{1-\kappa^2\sin^2\nu}}d\nu,\\ e^* &= 2E(\gamma,k) \equiv 2\int_0^{\gamma}\sqrt{1-\kappa^2\sin^2\nu}\,d\nu, \end{aligned} \tag{5}$$

and:

$$\begin{aligned} \gamma &= \sin^{-1}\sqrt{\frac{2\kappa}{\kappa+1}},\\ k &= \sqrt{\frac{\kappa+1}{2}}. \end{aligned} \tag{6}$$

In addition to the equations above, we can also apply an external force condition in that:

$$\alpha = \sqrt{\frac{P_0 l^2}{EI}}, \tag{7}$$

where $P_0 = p_0 A/M$. Here, p_0 is the global fluidic pressure, A is the Kirigami unit cell's area, and M is the number of cantilever beams in each layer ($M = 4$ in the square-cut design). In this way, the only unknown left in Eq. (4)–(7) is the bending moment M_0, which a simple numerical search can solve. Then, one can use the second and third equations in Eq. (4) to solve the beams' normalized tip displacement β_x and β_y.

The solutions above are formulated for a single nonlinear cantilever beam (or Kirigami ligament), but they apply to any ligaments in the square-cut Kirigami cell. Denote $\beta_y^{(1)}$ and $\beta_y^{(2)}$ as the normalized beam tip displacement of rings (1) and (2), respectively (the outermost ring is labeled as (1)). The overall Kirigami cell displacement Δ is:

$$\Delta = \beta_y^{(1)} l^{(1)} + \beta_y^{(2)} l^{(2)}. \tag{8}$$

This model's outcome is consistent with a previous work that only involves a single load P_0 [57]. To assess the precision of this model, we cut two Kirigami unit cells with different designs and measured their out-of-plane stiffness on a universal tensile test machine (0.19 mm thin color-coded plastic shim with $E \approx 5.5$ GPa and Instron 68TM-5 with 10N load cell). One unit cell sample is designed to be relatively stiff with $d = 50$ mm, $N = 1$, $w = 10$ mm, and $g = 5$ mm, while the other one is soft with $d = 80$ mm, $N = 2$, $w = 8$ mm, and $g = 7.5$ mm. They are both fixed to the tester table along its outer perimeter and subject to a point force at its middle via a 3D-printed customized adapter. Both Kirigami cells show consistent force–deformation curves through repeated loading cycles, and the test result agrees well with the analytical model prediction. Small discrepancies only occur when the external force is very high, when our $\theta_0 \approx 0$ assumption is no longer accurate. Regardless, this simple analytical model offers a powerful tool for designing the fluidic Kirigami morphing surface for accurate shape morphing.

2.2. Unique advantages of using fluidic Kirigami for shape morphing

The first advantage is its modularity. A single global pressure input can activate all the Kirigami unit cells. Moreover, cutting is a simple and scalable fabrication procedure (e.g., using a laser cutter or mechanical plotter cutter) [58,59]. Therefore, one can quickly design and cut Kirigami sheets on demand according to the desired morphing output. To demonstrate this, we cut two Kirigami sheets — each consisting of a uniform array of square-cut Kirigami unit cells — to "display" two patterns like a pixel grid. One is a smiley face, and the other pattern is the logo of the authors' institution (Fig. 2d). This morphing surface consists of a thin rubber sheet for pressure proofing, the plastic Kirigami sheet, and 6 mm thick aluminum support layer with square cutouts to prevent overall bulging from the pressurization (Section 2 of the Supplementary Materials and Movie S1 detail this test setup). With a global pressure input, each Kirigami cell will deform locally according to its underpinning design. Most importantly, if a new display pattern is desired, one can quickly swap Kirigami sheets with different designs.

The second advantage of the fluidic Kirigami morphing surface concept is its large design space. By simply tailoring the cutting geometry, we can tailor the unit cell deformation considerably. For example, Fig. 2e shows the result of a parametric study based on the aforementioned analytical tool. In this study, we fix the unit cell size $d = 9.5$ mm and the number of concentric rings $N = 1$ or 2. The actuation pressure is set at 60 kPa. By adjusting the cantilever beam width w and gap size g, one can continuously prescribe the unit cell deformation from 0 to 14 mm (a.k.a., **1.47** Δ/d aspect ratio). Therefore, it becomes possible to fine-tune the deformation of each unit cell accurately and independently within a large range.

2.3. Demonstrating high-precision shape morphing with ultrasonic holographic lensing

To exemplify these two advantages of fluidic Kirigami morphing surface for high-precision shape morphing, we apply this concept and create a programmable ultrasonic holographic lens (Fig. 3). Acoustic

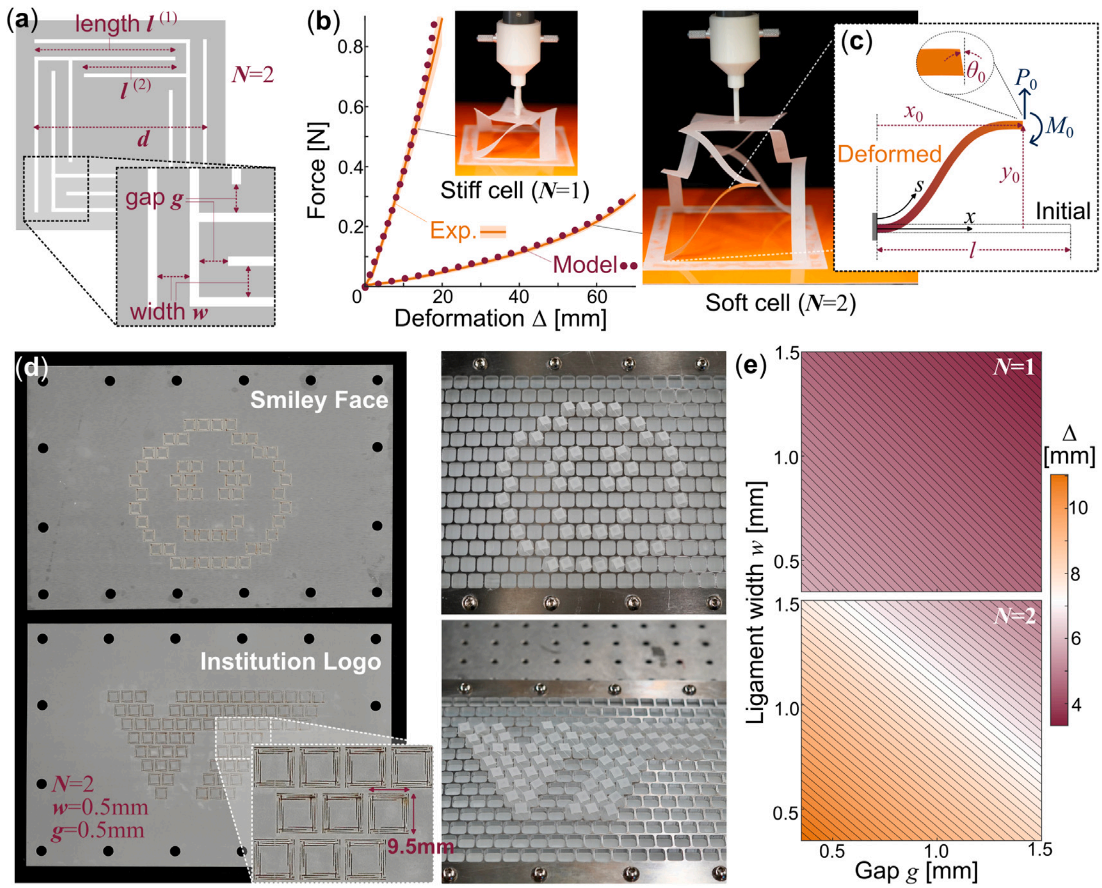

Fig. 2. The anatomy and design space of a square-cut fluidic Kirigami morphing surface. (a) An example design with two concentric rings of straight ligaments (N = 2). Here, we use superscript $^{(N)}$ to denote the different rings. (b) The out-of-plane deformation response of two square-cut Kirigami cells. The solid lines are averaged test results from 10 consecutive loading cycles, and the shaded bands are the corresponding standard deviation. The dotted lines are the analytical model prediction. (c) The deformation kinematics of an individual ligament modeled as a nonlinear Euler–Bernoulli beam. (d) A demonstration of pre-programmed Kirigami sheets "displaying" two different patterns. One can simply swap a new Kirigami sheet if a new pattern is desired. (e) A parametric study showing the design space of 9.5 mm × 9.5 mm square-cut Kirigami unit cells subject to 60 kPa fluidic pressure, based on the analytical solution defined in Eq. (8).

holography is a technique to reconstruct a three-dimensional sound field and visualize sound propagation by leveraging wave interference [60–64]. Such holography can be achieved by specialized lenses, which have a carefully defined and non-smooth thickness map. As the sound wave front travels through or reflects from the lens, this complex distribution of thickness will modulate the wave front's phase and amplitude [65]. As a result, the acoustic wave energy will be focused into pre-defined patterns with diffraction-limited resolution at the target plane [60] (Fig. 3a). Typically, a holographic lens is entirely static (e.g., a 3D-printed disc [66–68]) or entirely active (e.g., phased array transducers (PATs) [69–71]). The static ones are limited to one particular holographic pattern, while the active ones come with a high cost, especially with high-frequency applications [72–75]. The fluidic Kirigami holographic lens can strike a balance between cost and versatility.

As a demonstration, we designed a fluidic Kirigami holographic lens that can steer plane acoustic wavefronts and focus them into two distant points at the target plane (Fig. 3b). To design such a lens, one can integrate the parametric design space as shown in Fig. 2e and the Iterative Angular Spectrum Approach (IASA) [76–78]. The IASA typically starts from an initial holographic lens design and iteratively updates this design by repeating five consecutive steps. (1) Propagating the acoustic field from the initial lens to the target plane — using numeric simulation — and evaluating the quality of the projected acoustic "image". (2) Imposing the desired amplitude pattern on the target plane (e.g., the two focus points in our example) while retaining the phase. (3) Propagating the adjusted acoustic field from the target plane back to the hologram plane. (4) Computing the lens thickness map. (5) Updating the complex amplitude at the hologram plane accordingly and repeating the steps.

We first conducted a characterization experiment to obtain the source complex amplitude without the Kirigami sheet (a flat surface). This result was then treated as the "initial lens design" for obtaining the desired thickness map using IASA. We assume a 1D propagation within the structure, neglecting shear waves and attenuation. The transmission losses are considered minimal and, therefore, also neglected. Fig. 3b presents the simulated wave field amplitude at the target plane, and Fig. 3c shows the final thickness map targets from the IASA. Then, we can simply find the corresponding Kirigami cut design by matching the target thickness with the parametric design space in Fig. 2e. It is noted that, with the available number of Kirigami cells (52 in our case), the image accuracy is not optimal. The pressure at the center, generated by the piezoelectric disk, is not entirely eliminated and directed toward the two points. This issue could be mitigated by increasing the operating frequency, thereby increasing the total number of cells. However, designing smaller cells while maintaining similar levels of out-of-plane thickness poses a significant fabrication challenge and is beyond the scope of this work. Section 3 of the Supplement Materials derive the ultrasonic frequency limit using current fabrication setup.

Fig. 3e–f shows the final fluidic Kirigami lens deformation under global pressure and the experimentally measured pressure amplitude and phase distribution measured by the hydrophone (Section 4 of the Supplementary Materials details the experimental setup). The activated state of the Kirigami cells leads to the formation of a hologram-like

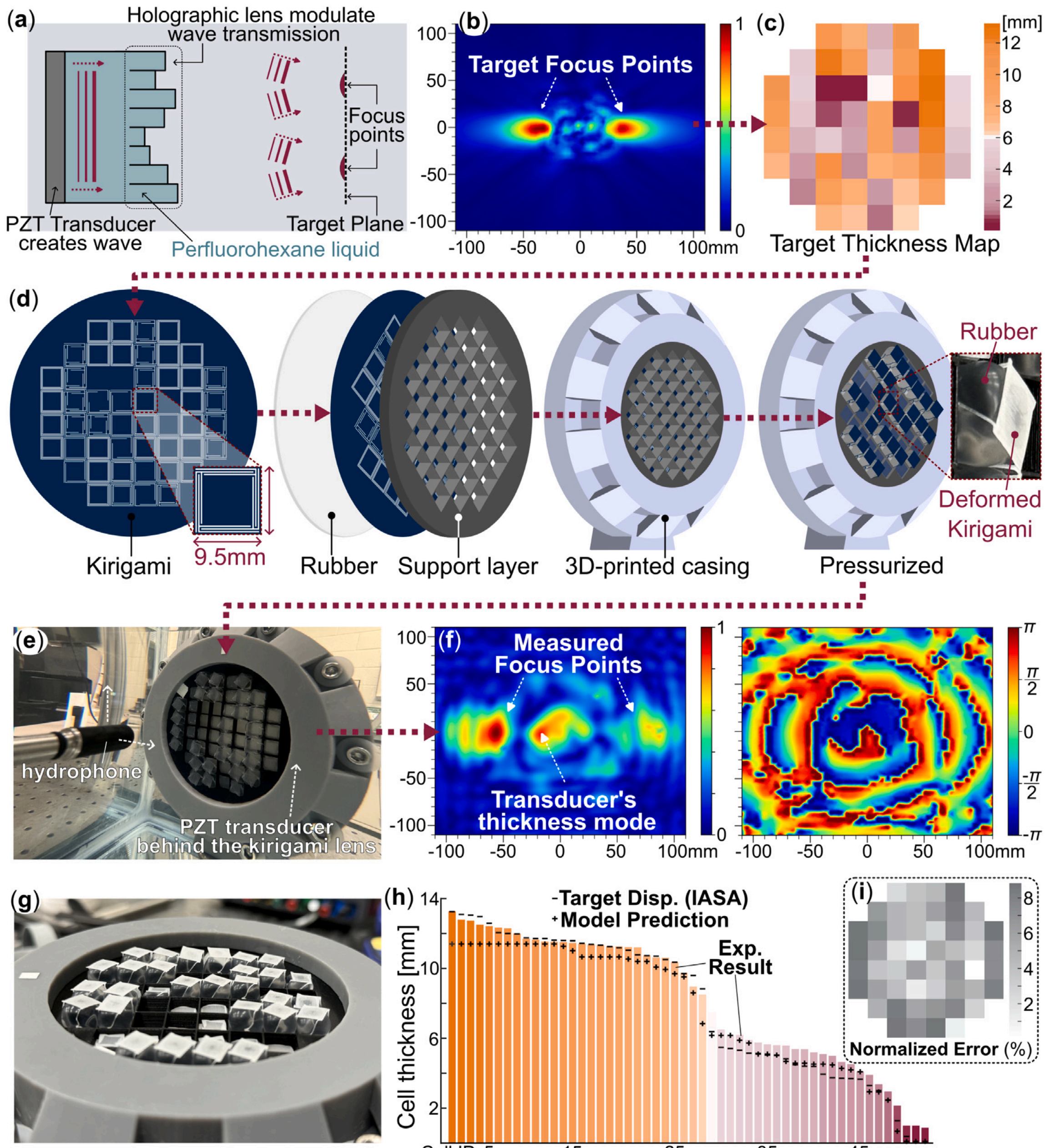

Fig. 3. The making and testing of a fluidic Kirigami holographic lens for ultrasonic waves. (a) A schematic diagram showing the fundamental working principle of an acoustic holographic lens. Notice that an accurate thickness distribution of the holographic lens is crucial for steering the acoustic waves. (b) The targeted two focus points in this demonstration. (c) The desired lens thickness map created by the IASA method, involving 52 cells of 9.5 mm × 9.5 mm size. (d) From left to right: The Kirigami sheet is designed by matching the target lens thickness map from IASA and the design space in Fig. 2e → then this Kirigami is sandwiched between a 1 mm thin silicone rubber sheet and 6 mm thin 3D-printed PLA support layer → then everything is encapsulated in 3D-resin-printed casing filled with working liquid inside (perfluorohexane) → finally, upon pressurization, the Kirigami cells will morph into a holographic lens with a correct thickness map. (e) The fluidic Kirigami holographic lens under testing. The whole setup is submerged underwater. (f) The measured acoustic pressure at the target plane and the corresponding phase distribution. The two focus points are evident. Note that the phase information is not used in IASA method, so it is displayed only for reference. (g) A close-up of the fluidic Kirigami ultrasonic lens under pressure. (h) Comparing the actual thickness map with the targeted values. For visual clarity, we re-arrange the 52 cells according to their out-of-plane thickness. The bars are measured unit cell thickness using an optical sensor (optoNCDT 2300-50 laser triangulation sensors, Micro-Epsilon), "−" denote the targeted thickness from the IASA method, and "+ " are the analytical model predictions. (i) The corresponding thickness error map as a percentage of acoustic wave length at 70 kHz.

metasurface (Fig. 3e), which introduces a phase offset for the transmitted acoustic wave to achieve the desired pressure fields — a two-point pattern in this case (Fig. 3f).

Finally, Fig. 3(g–i) highlights that the displacement of each Kirigami cell agrees well with the targeted thickness map despite the additional rubber sheet, which is not modeled in our theory. Fig. 3(h) compares three sets of data: They are (1) the targeted unit cell thickness map based on the IASA method, (2) the predicted unit cell thickness based on the nonlinear mechanics model, and (3) the experimentally measured unit cell thickness. For visual clarity, we arrange the unit cells based on their thickness, rather than their spatial locations in the morphing surface. Fig. 3(i) is the extension of 3(g), where we calculate and map the percentage error between the targeted and actual unit cell thicknesses. Since this setup functions as a holographic metasurface,

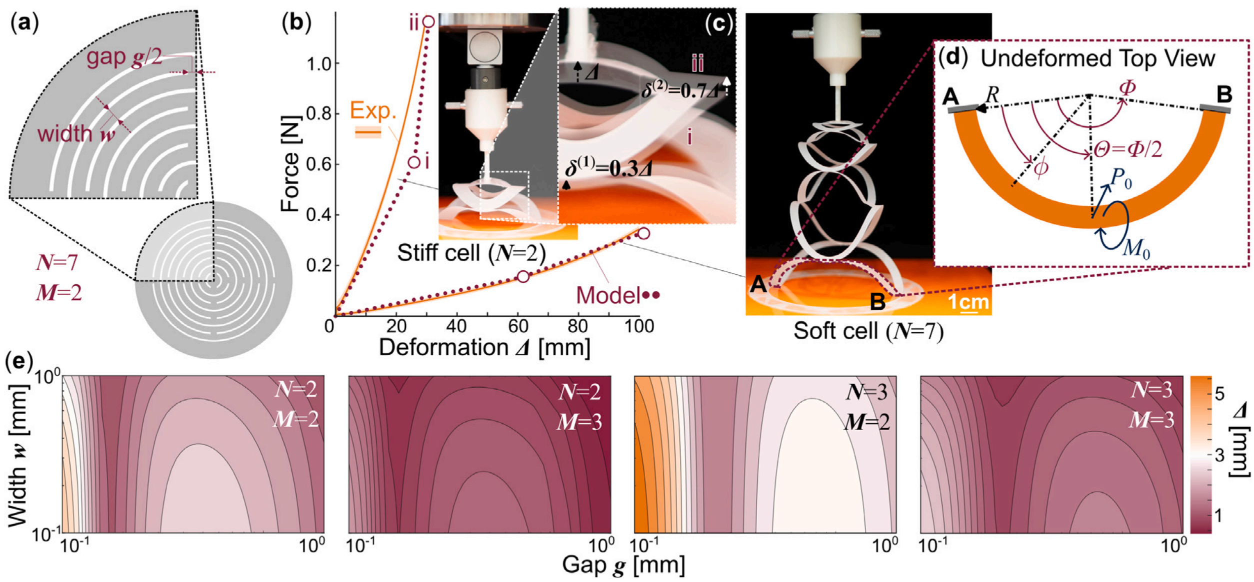

Fig. 4. The anatomy and design space of circular-cut Kirigami (a) An example design of a circular-cut Kirigami cell consisting of 7 concentric rings of cuts ($N = 7$), with each ring having 2 interconnected ligaments ($M = 2$). Again, we label the outermost ring by superscript $^{(1)}$. (b) The out-of-plane force–deformation relationships of soft and stiff circular Kirigami cells. Notice that the model predictions (maroon dotted line) are only piecewise linear, but they agree reasonably well with experiment data (orange solid line). (c) A composite image comparing the Kirigami deformation at two different deformation stages. At the high deformation range, the inner ring deforms much more significantly than the outer ring. This is consistent with the piecewise linear model assumption. (d) Kinematics of an undeformed ligament. (e) The modeled out-of-plane deformation corresponding to different circular Kirigami designs.

the associated error in cell displacements must not exceed the sub-wavelength threshold ($\lambda/10$) of the surrounding medium (water). This corresponds to an allowable error of less than 2.11 mm at 70 kHz for each Kirigami cell. The fluidic Kirigami morphing metasurface successfully reaches this tight tolerance with ≤2 mm error, indicating its capability to achieve high-precision morphing despite the additional rubber sheet beneath.

However, despite the very accurate shape morphing from the Kirigami metasurface, the pressure fields observed in the experiments do not perfectly align with the simulation data (i.e. comparing Fig. 3b and f). The primary source of this error is the relatively low spatial resolution of the fluidic origami unit cells and, consequently, the relatively low working frequency. Therefore, reducing the unit cell size and increasing the working frequency to the lower megahertz scale is the most effective strategy to replicate the two-point focus pattern accurately. However, this would require new micro-fabrication techniques, as we discussed earlier, and will be the topic of future work.

There are also additional, albeit minor, sources of error, including the effect of gravity, differences in hydrostatic pressure on the Kirigami cells due to the interaction of the two fluids (water and perfluorohexane) at their surfaces, and the bulging of the rubber sheet through the Kirigami sheets. Most notably, the IASA simulation assumes that each cell has a perfect cuboid shape with sharp edges, but the inflated fluidic Kirigami cells have a more dome-like shape.

All these primary and minor sources of error accumulate and lead to inaccuracies in directing the acoustic energy to the two focus points and result in the unintended preservation of the natural thickness mode of the transducer, which appears as an additional point at the center between the two focal dots (Fig. 3f). Nonetheless, our experiment results successfully proved the fundamental working principle of holographic lensing with fluidic Kirigami.

3. Circular-cut Kirigami surface for high-resolution force control

When viewed from the top, a circular-cut Kirigami cell consists of multiple concentric "rings", each containing several thin and interconnected curved ligaments (or beams). Fig. 4a–d highlights one such curved ligament, which connects to other ligaments *in the same layer* at its ends and connects to the ligaments *in the inner layer* at its center.

3.1. Analytical tool for analyzing the circular cut Kirigami cells' deformation

Unfortunately, analyzing the out-of-plane deformation of these curved structures becomes much more complicated than the simple straight ligaments in the square-cut Kirigami; solving the nonlinear Euler–Bernoulli equation on a curved beam becomes mathematically intractable. To this end, we observe that — as the circular Kirigami cells undergo out-of-plane deformation — the outer rings of curved ligaments deform more significantly than the inner rings because the former have slender shapes. As a result, the outer rings will reach plastic yielding first. Therefore, *we proposed a novel approach by approximating the large amplitude and nonlinear out-of-plane deformation of the circular-cut Kirigami with a piecewise linear model*, so that:

$$\Delta = \sum \delta^{(i)}, \text{ where } \delta^{(i)} = \begin{cases} \delta^{(i)}(p_0), & \text{before elastic limit } (S_V^{(i)} \le S_{yield}), \\ \delta^{(i)}_{\max}, & \text{beyond elastic limit } (S_V^{(i)} > S_{yield}). \end{cases} \tag{9}$$

Here, p_0 is the pressure input. $\delta^{(i)}(p_0)$ is the curved beam's vertical deformation before reaching the elastic limit, correlating linearly to p_0 (again, the superscript $^{(i)}$ explains in which ring this curved ligament is located). Meanwhile, $\delta^{(i)}_{\max}$ is a constant value corresponding to the ligament deformation beyond the elastic limit. $S_V^{(i)}$ and S_{yield} is the Von Mises stress and yield stress, respectively. As the circular-cut unit cell deforms under pressure, its outer rings of thin beams deform more significantly than the inner layers because the former have more slender shapes. As a result, these outer layers will reach an "elastic limit" first (more on this later), creating a nonlinear behavior even though the beam deformation follows linear mechanics. In what follows, we detail these two deformation stages for a curved beam (while dropping the superscript $^{(i)}$ for clarity).

Before elastic limit: According to *Castigliano's second theorem*, the deformation of a beam is governed by:

$$\delta = \frac{\partial U}{\partial Q}, \tag{10}$$

where U is the strain energy and Q and δ are the generalized force and displacement, correspondingly. While the curved beam deformation in the Kirigami cell can be complex, careful observation of the experiment results suggests that it predominantly bends out-of-plane without significant twisting, shear, or stretching deformations (Fig. 4b,c). In addition, the geometry and loading of the Kirigami cell are axisymmetric. Therefore, we can assume each curved beam is fixed at both ends and subjected to two external loads at its center: an out-of-plane force P and a twisting moment (torque) T in a plane normal the beam's curved axis.

Therefore, denote ϕ as the angular position along the curved beam's span, and Φ is the beam arc's total span ($\phi \in [0 \ldots \Phi]$) (Fig. 4d). The strain energy is given by [79]:

$$U \simeq U_f = \int_0^l \frac{M(\phi)^2}{2EI}\, ds = \int_0^\Phi \frac{M(\phi)^2}{2EI} R\, d\phi, \tag{11}$$

where U_f is the complementary flexure energy, EI is the flexural rigidity along the bending axis, R is the curved beam's radius, and l is the beam length. Since there are two external loads (P and T), the reaction bending moment $M(\phi)$ at position ϕ consists of two components in that [79]:

$$\begin{aligned} M(\phi)_P &= V_{A,P} R \sin\phi + M_{A,P}\cos\phi - T_{A,P}\sin\phi - PR\sin(\phi-\Theta)\langle\phi-\Theta\rangle^0, \\ M(\phi)_T &= V_{A,T} R \sin\phi + M_{A,T}\cos\phi - T_{A,T}\sin\phi - T\sin(\phi-\Theta)\langle\phi-\Theta\rangle^0, \end{aligned} \tag{12}$$

where V_A, M_A, and T_A are the reaction force, bending moment, and twisting moment at the curved beam's left end A, respectively. Θ is the external loads' position, and in this case, $\Theta = \Phi/2$. The angle brackets $\langle\phi-\Theta\rangle^n$ are defined such that $\langle\phi-\Theta\rangle^n = 0$ when $\phi < \Theta$, and $\langle\phi-\Theta\rangle^n = (\phi-\Theta)^n$ otherwise. Based on the formulations above, the work of [79] showed that the reaction forces and moments at the end of the beam are linearly correlated with the external load in that:

$$\begin{aligned} &V_{A,P} = \alpha_V P,\ M_{A,P} = \alpha_M PR,\ T_{A,P} = \alpha_T PR, \\ &V_{A,T} = \beta_V \frac{T}{R},\ M_{A,T} = \beta_M T,\ T_{A,T} = \beta_T T, \end{aligned} \tag{13}$$

where α and β are non-dimensional constants defined by the curved beam's geometry and consecutive properties. Detailed derivations of these constants are available in Section 5 of the Supplementary Materials. The curved beam's out-of-plane deformation becomes:

$$\delta = \delta_P - \delta_T. \tag{14}$$

Notice the negative sign indicates that the twisting moment opposes the out-of-plane force. Moreover:

$$\begin{aligned} \delta_P &= M_{A,P}\frac{R^2F_1}{EI} + T_{A,P}\frac{R^2F_2}{EI} + V_{A,P}\frac{R^3F_3}{EI} - P\frac{R^3F_{a3}}{EI}, \\ \delta_T &= M_{A,T}\frac{R^2F_1}{EI} + T_{A,T}\frac{R^2F_2}{EI} + V_{A,T}\frac{R^3F_3}{EI} + T\frac{R^2F_{a2}}{EI}. \end{aligned} \tag{15}$$

Here, F_i are also non-dimensional constants detailed in Section 5 of the Supplementary Materials. We assume the contact between the pressurized rubber sheet and the Kirigami sheet occurs primarily at the center of its cells, allowing us to approximate the overall out-of-plane force as a concentrated point load at the cell center. Therefore, in the fluidic morphing surface, $P = p_0 A/M$, where p_0 is the global fluidic pressure, A is the unit cell's area in contact with the Kirigami sheet, and M is the number of curved beams in each layer. The twisting moment is $T = p_0 A \times R/M$. Eqs. (14) and (15) give the linear deformation mechanics as described in the first line of Eq. (9).

Beyond elastic limit: While Eq. (15) describes the linear elastic deformation of the curved beam, we observe that a nonlinearity occurs when the outer rings reach their elastic limit before the inner rings. Here, we define the elastic limit based on yield stress. When the bending stress S_{M_A} reaches the yield stress S_{yield} at the beam's fixed end (where the bending stress is maximum), the elastic limit is reached. As shown in Fig. 4c, it deforms much slower after the outer layer of curved beams reaches the elastic limit. Meanwhile, the inner layers continue to bend. Therefore, we assume the curved beam deflection is constant after the limit is reached (i.e., the second line of Eq. (9)). The *Von Mises stress* S_V is:

$$S_V = \sqrt{S_{M_A}^2 + 3\left(S_{T_A} + S_{V_A}\right)^2} \simeq S_{M_A} = \frac{M_A c}{I}, \tag{16}$$

where I is the area moment of inertia, and $c = t/2$ is the distance between the beam's neutral axis and its surface. Since in the yield region $S_V = S_{\text{yield}}$, the maximum reaction bending moment is:

$$M_{A,\max} = \frac{S_{\text{yield}} I}{c}. \tag{17}$$

Based on the tensile test results, $S_{yield} \approx 160$ MPa. Using the second and fourth equations in Eq. (13), we can solve for $P_{\max}$ and $T_{\max}$, representing the external loads at the elastic limit. We then use the rest of Eq. (13) to solve for the corresponding reaction force and moments $V_{A,P_{\max}}$, $T_{A,P_{\max}}$, $V_{A,T_{\max}}$, and $T_{A,T_{\max}}$. Finally, we can plug in these maximum reaction forces and moments into Eqs. (14), (15) to obtain the curved beam deformation at the elastic limit $\delta_{\max}$.

We tested the accuracy of this piecewise linear model on two unit cell samples (Fig. 4b). One cell is relatively stiff with $N = 2$, $M = 2$, $d = 70$ mm, $w = 10$ mm, and $g = 10$ mm, while the other cell is softer with $N = 7$, $M = 2$, $d = 80$ mm, $w = 5$ mm, and $g = 5$ mm. Certainly, the piecewise linear model is not as accurate as the fully nonlinear model discussed earlier, but it still provides a reasonable prediction without unnecessary mathematical complexity.

Moreover, based on this validated analytical model, one can also explore the circular-cut Kirigami's design space for prescribing its out-of-plane deformation. Fig. 4e summarizes the results of such a parametric study where the unit cell size is $d = 9.5$ mm, the number of concentric rings varies from $N = 2$ to $N = 3$, and the number of curved ligaments in each ring varies from $M = 2$ to $M = 3$. The actuation pressure is set at 55 kPa. By adjusting the curved beam's width w and gap size g, one can continuously prescribe the unit cell deformation from 0 to 6 mm (i.e., 0.67 Δ/d aspect ratio). Moreover, the design mapping is no longer 1-to-1. There can be multiple design choices for a targeted out-of-plane deformation Δ, giving us the flexibility to consider other design requirements (as we detail in the example below).

3.2. Using fluidic Kirigami for high-resolution force control in a haptic device

Haptic devices for human touch-related neurophysiological investigations typically apply a high-resolution, low magnitude force by relying on kinematic (rigid) mechanisms or piezoelectric materials [80–82]. Another approach that is becoming more common is to use a fluid (e.g., pneumatic system that provides air) to apply forces, including via a flexible membrane (Fig. 5) [83–86]. Yet, this latter approach can result in poor force control and estimation for reasons including a changing contact area and uneven indentation depth [84]. Integrating a Kirigami layer can stabilize the contact area and allow more uniform deformation, addressing the aforementioned limitations of this fluidic control approach.

Here, we present an example pneumatically-controlled haptic device designed for human touch-related neurophysiological investigations that employs a circular-cut Kirigami mechanics model. A comprehensive characterization of this device, which can apply, modify, and measure forces that are small in amplitude (e.g., 0.01N), is provided in Kalantaryardebily et al. [87]. We use this example to show how to achieve a desired elastic force and displacement range by adjusting the stiffness generated by the Kirigami layer. Compared with our previous prototype, which was used in touch perception investigations [84,88], this device provides enhanced real-time force estimation and more precise force application, both enabled by integrating the Kirigami layer [87].

In this example, we aimed to create a compact haptic "button" that can deliver forces in the range of 0–4 N and displacements in the

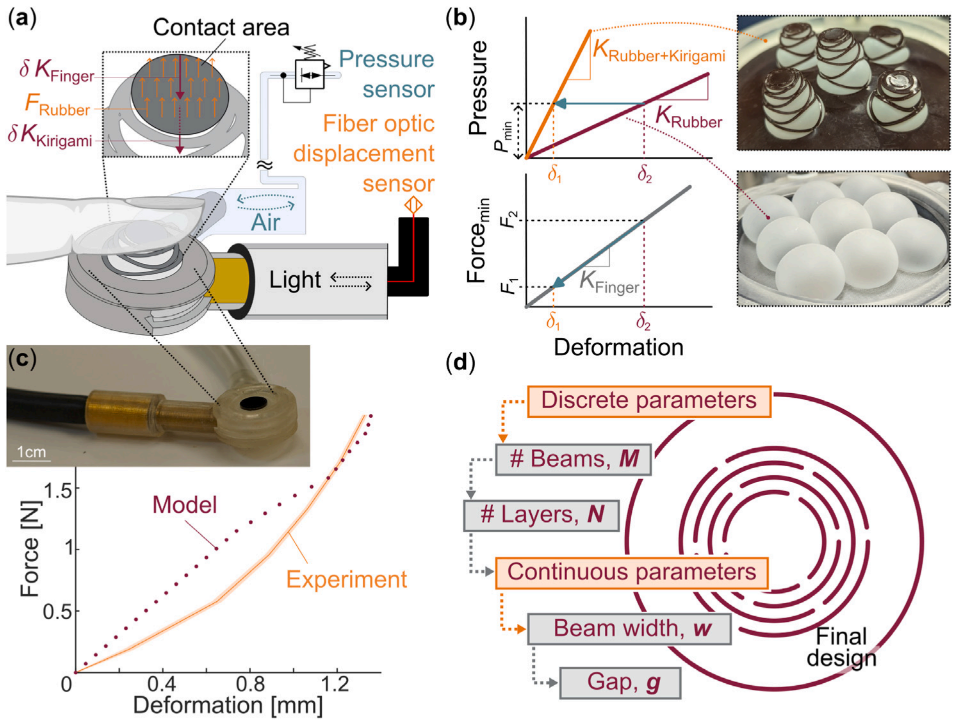

Fig. 5. Designing fluidic Kirigami for haptic interfacing (a) This haptic device is designed to apply a precise and low-amplitude force stimulus — via a fixed contact area — to an individual's finger, given the primary goal of investigating how such forces are detected by the individual. The device incorporates two sensors: a pressure sensor and a fiber optic displacement sensor. When the force stimulator makes contact with the individual's finger, the displacement of the Kirigami and finger are the same (denoted as δ). This displacement results in a force being applied. The displacement is measured using a fiber optic displacement sensor, and, combined with the pressure measured from the pressure sensor, is used to estimate the applied force. K_{Finger} and $K_{Kirigami}$ are the stiffness of the finger and Kirigami layer, respectively, and F_{Rubber} is the pressure force exerted by the sealing rubber layer. The magnitude of the force stimulus can be adjusted by modulating the fluidic air pressure using a pneumatic control system. The cut pattern design of the Kirigami layer dictates the force–deformation relationship, which can be programmed with our piecewise linear model. (b) Here, we use illustrative force–deformation curves with and without the Kirigami layer to explain the benefit of fluidic Kirigami. Due to the stiffness increase from Kirigami, the finger displacement decreases from δ_2 to δ_1 for the minimum input pressure increment. As a result, the force transmitted to the finger decreases from F_2 to F_1, thus providing a higher-resolution stimulus. (c) The experimentally measured free-stroke force and displacement relationship agrees reasonably well with the theoretical prediction. For the experimental data, the fiber optic sensor tracked the displacement of the Kirigami's contact area, as shown in (a), and the pressure sensor helped estimate the force applied to the Kirigami layer. The solid line is the averaged data from 5 loading/unloading cycles, and the thin shadow is the corresponding standard deviation, showing highly repeatable performance. The discrepancy in this figure is due to the fundamental inaccuracy of the piecewise linear model used for the circular cut pattern, which did not practically limit this project. (d) Design flow chart for the Kirigami to enable a high-resolution force output. The discrete parameters of the number of beams (M) and number of rings (N) allow for greater changes in the force–displacement curve, whereas the continuous parameters of the beam width (w) and gap (g) allow for fine-tuning.

range of 0–1.5 mm, with force increments lower than the threshold of human force detection (0.05N) (Fig. 5a) [82,89]. One of the key considerations here is force reproducibility, repeatedly delivering the same low-amplitude force across many stimuli [82,90]. The deformation must remain within the elastic region to allow this repeatable force application. Our previously discussed piecewise linear model can readily predict the applied force and expected displacement while remaining within the elastic deformation limit (Fig. 5c), enabling the design of an appropriate cut tailored for this specific application.

Our piecewise linear model highlights four key geometrical variables that govern the force–displacement response of a circular-cut Kirigami cell: the number of "rings" defined by the concentric cuts N, the number of curved beams in such a ring M, the width of these curved beams w, and the gap between adjacent cut tips g (Fig. 4a). In principle, one can achieve the desired force–deformation output by looking up the design space in Fig. 4e. However, the order of selecting these design parameters is critical to ensure that the fluidic Kirigami remains within the elastic region. Because stress concentrates near the tips of slit cuts, the number of curved beams in each ring M has the most significant influence on the elastic deformation limit. Additionally, the number of rings N strongly constrains the overall magnitude of the elastic deformation range. Therefore, we first selected M, followed by N, using the design space in Fig. 4e. Next, we selected the curved beam width w, followed by the cut gap g to fine-tune the fluidic Kirigami's mechanical response. Unlike M and N, the final two variables are continuous, offering considerable design freedom. Fig. 5d summarizes the overall design flow chart.

For the example haptic device shown here, we chose three beams in each ring ($M = 3$) to ensure an elastic response within the desired force/deformation range (Fig. 5c). The number of rings, N, was set at 3 to guarantee sufficient deformation while ensuring easy fabrication of the Kirigami layer. The gap and beam width were set at 0.7 mm and 0.7 mm, respectively, to achieve the targeted 0–4 N output force. Fig. 5c shows the final prototype based on the aforementioned fluidic Kirigami design (more haptic experiment details are available in Section 6 and Video S2 of the Supplementary Materials). The experimentally measured output force and displacement generally aligns with the theoretical predictions, with both remaining within the elastic region. Discrepancy exists in the mid-range due to the inherent limitation of the piecewise linear modeling assumptions, but the model is still sufficient to guide the Kirigami design. Overall, this Kirigami layer allows for repeatable deformation when the input pressure is low, achieving the modeled force–displacement curve.

An added benefit of including the Kirigami layer is that the increased stiffness when deforming out-of-plane permits higher resolution in force control. This benefit arises particularly when the stiffness of the rubber sheet is small in comparison to the combined Kirigami and rubber layer. As shown in Fig. 5b, the stiffness of the combined rubber and Kirigami layer is greater than of the rubber layer alone. Due to the nature of the pressure supply (i.e., pressure regulator), there is a minimal increment of internal pressure P_{min}. For a high-performance haptic device, one would expect the increase in contact force to be small from such a minimal pressure increment. Assume the haptic device is already in complete contact with the individual's finger; as a result, the displacements of the haptic device's contact surface and the finger tip are the same. Suppose the haptic device does not have the Kirigami layer (i.e., sealing rubber only). The haptic device's displacement is only related to the rubber's stiffness, resulting in a relatively large deflection δ_2. As a result, the increase in the haptic contact force due to P_{min} is also larger at $F_2 = K_{\text{Finger}} \times \delta_2$. However, when the Kirigami layer is present, the haptic device's stiffness becomes significantly higher. As a result, the contact surface's displacement becomes smaller ($\delta_1 < \delta_2$), and the increase in haptic contact force F_1 also becomes smaller. Therefore, the force resolution of the stimulus improves.

Another benefit of the Kirigami layer is its safety and functionality as a haptic device. When inflating with a large pressure, the hardware (e.g., the rubber sheet) will fail due to bursting and/or breakdown. The additional stiffness from the Kirigami layer will increase the failure pressure, thereby increasing the maximum usable force.

While we focus primarily on the force control capabilities, we also would like to highlight the benefits of the Kirigami layer for sensing. The cut-free center region provides a mounting surface for a reflecting film used with a sensor (e.g., fiber optic sensor), allowing accurate displacement measurement and better force estimation. This accurate measurement would not be possible if only the rubber sheet was included, since inflation could occur in varying directions.

Finally, we note that the same design principle presented here could be applied to other haptic areas of interest that require precise and adaptive force control. By modifying the theoretically-identified design parameters, the haptic device can be created for use in areas including virtual reality systems for one-dimensional force feedback or robotic surgery. Furthermore, future designs can consider how to adapt the circular- and square-design theory presented here to create haptic devices that permit sensing and actuation in multiple degrees-of-freedom.

4. Discussion

In this study, we developed a fluidic Kirigami morphing surface capable of precise shape morphing and control of the output force, leveraging geometric design principles rather than complex material engineering. Through theoretical modeling and experimental validation, we demonstrated how square and circular-cut Kirigami cells enable tunable morphing and how to apply them to different multiphysics applications like ultrasonic holography (high-precision out-of-plane deformation) and haptic interfaces (high-resolution out-of-plane force control). Our results show that fluidic Kirigami morphing surface provides a scalable, easily manufacturable solution for morphing surfaces, as cutting is an inherently accurate and scalable fabrication method.

Another notable contribution of this study is the new nonlinear mechanics model of the square and circular-cut Kirigami cell. We adopted two approaches to analyze the Kirigami deformation according to their design: A fully nonlinear Euler beam approach for the straight ligaments in the square-cut Kirigami and a piecewise-linear approach for the curved ligaments in the circular-cut Kirigami. Certainly, this model cannot be used for every Kirigami surface. The design and deformation possibilities of Kirigami is seemly infinite. However, our models are, in principle, applicable to a large subset of Kirigami designs as long as they satisfy two criteria: (1) the Kirigami cutting pattern divides the thin-sheet precursor material into a network of interconnected and narrow ligaments, and (2) these ligaments are subject to out-of-plane bending (that is, one deforms the Kirigami sheets out-of-plane rather than stretching them in-plane). Based on these two criteria, either of these two modeling approaches (or a combination of them) could apply to other Kirigami designs.

It is also worth re-iterating that, although the silicone sealing layer plays an important role in the fluidic Kirigami, we found that it is not crucial to include explicitly in our mechanics model. This is because the stiffness of the Kirigami layer dominates that of the silicone layer, as well as the experimental implementation in this study. In the holographic lensing demonstration, pressure control is open-loop. We manually and slowly increase the global pressure until the desired thickness distribution among the fluidic Kirigami unit cells is achieved. The model-predicted distribution of the unit cell displacement is very close to the experimental data (Fig. 3h), suggesting that the silicone layer plays a minor role in fluidic Kirigami's out-of-plane displacement. In the haptic interface demonstration, the fluidic Kirigami also operates with an open-loop control (except for the input pressure, which is provided by a pressure regulator with a closed-loop pressure feedback). We only need to use the mechanics model to design the reaction force provided by the Kirigami layer.

Regarding future work, the fluidic Kirigami morphing surface concept could benefit from more advanced micro-fabrication methods. This would enable smaller and denser unit cells for new functions (e.g., pushing the usable frequency of ultrasonic holographic lensing to MHz). Including rubber sheets in the mechanics, model will further improve modal accuracy. Finally, directly embedding the Kirigami layer inside the rubber sheet and eliminating the friction contact between them could allow for a more consistent output at higher pressure. Regarding haptic interfacing, it is also necessary to conduct extensive studies to further characterize the fluidic Kirigami's mechanical response, such as displacement and force resolution, hysteresis and repeatability, and force output under dynamic pressure. To this end, we have conducted a parallel study [87]. Lastly, by refining the control and scalability of these morphing surfaces, this study lays the foundation for next-generation adaptive materials with broad applications in acoustic manipulation, haptic feedback, and reconfigurable structures.

CRediT authorship contribution statement

Ardalan Kahak: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Moustafa Sayed Ahmed:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Nahid Kalantaryardebily:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Hrishikesh Kulkarni:** Writing – original draft, Visualization, Data curation. **Netta Gurari:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Shima Shahab:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Suyi Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eml.2025.102424.

Data availability

Data will be made available on request.

References

[1] J. Wang, A. Chortos, Performance metrics for shape-morphing devices, Nat. Rev. Mater. 9 (10) (2024) 738–751.
[2] X. Yang, Y. Zhou, H. Zhao, W. Huang, Y. Wang, K.J. Hsia, M. Liu, Morphing matter: From mechanical principles to robotic applications, Soft Sci. 3 (4) (2023) N–A.
[3] C. Thill, J. Etches, I. Bond, K. Potter, P. Weaver, Morphing skins, Aeronaut. J. 112 (1129) (2008) 117–139.
[4] S. Barbarino, O. Bilgen, R.M. Ajaj, M.I. Friswell, D.J. Inman, A review of morphing aircraft, J. Intell. Mater. Syst. Struct. 22 (9) (2011) 823–877.
[5] E. Ajanic, M. Feroskhan, S. Mintchev, F. Noca, D. Floreano, Bioinspired wing and tail morphing extends drone flight capabilities, Sci. Robotics 5 (47) (2020) eabc2897.
[6] B. Johnson, M. Naris, V. Sundaram, A. Volchko, K. Ly, S. Mitchell, E. Acome, N. Kellaris, C. Keplinger, N. Correll, et al., A multifunctional soft robotic shape display with high-speed actuation, sensing, and control, Nat. Commun. 14 (1) (2023) 4516.
[7] I.P. Qamar, R. Groh, D. Holman, A. Roudaut, HCI meets material science: A literature review of morphing materials for the design of shape-changing interfaces, in: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, 2018, pp. 1–23.
[8] Y. Bai, H. Wang, Y. Xue, Y. Pan, J.-T. Kim, X. Ni, T.-L. Liu, Y. Yang, M. Han, Y. Huang, et al., A dynamically reprogrammable surface with self-evolving shape morphing, Nature 609 (7928) (2022) 701–708.
[9] J.H. Pikul, S. Li, H. Bai, R.T. Hanlon, I. Cohen, R.F. Shepherd, Stretchable surfaces with programmable 3D texture morphing for synthetic camouflaging skins, Science 358 (6360) (2017) 210–214.
[10] M. Murashima, Y. Imaizumi, R. Murase, N. Umehara, T. Tokoroyama, T. Saito, M. Takeshima, Active friction control in lubrication condition using novel metal morphing surface, Tribol. Int. 156 (2021) 106827.
[11] S. Venkatesh, D. Sturm, X. Lu, R.J. Lang, K. Sengupta, Origami microwave imaging array: Metasurface tiles on a shape-morphing surface for reconfigurable computational imaging, Adv. Sci. 9 (28) (2022) 2105016.
[12] B. Assouar, B. Liang, Y. Wu, Y. Li, J.-C. Cheng, Y. Jing, Acoustic metasurfaces, Nat. Rev. Mater. 3 (12) (2018) 460–472.
[13] C. Zhang, W.K. Cao, L.T. Wu, J.C. Ke, Y. Jing, T.J. Cui, Q. Cheng, A reconfigurable active acoustic metalens, Appl. Phys. Lett. 118 (13) (2021).
[14] Z. Tian, C. Shen, J. Li, E. Reit, Y. Gu, H. Fu, S.A. Cummer, T.J. Huang, Programmable acoustic metasurfaces, Adv. Funct. Mater. 29 (13) (2019) 1808489.
[15] W.K. Cao, C. Zhang, L.T. Wu, K.Q. Guo, J.C. Ke, T.J. Cui, Q. Cheng, Tunable acoustic metasurface for three-dimensional wave manipulations, Phys. Rev. Appl. 15 (2) (2021) 024026.
[16] Z. Zhu, S. Cheng, J. Han, S. Yan, P. Fan, Q. Hu, Z.-H. Tang, Electrically reconfigurable metasurfaces for frequency selective transmission via IPMC Kirigami, Adv. Mater. Technol. 9 (7) (2024) 2301879.
[17] H. Luo, Y. Wang, M. Wang, C. Wang, T. Liu, X. Ling, H.-X. Xu, Reconfigurable high-efficiency metadevice using Kirigami-inspired phase gradient metasurfaces, Results Phys. 53 (2023) 106949.
[18] H.-X. Xu, M. Wang, G. Hu, S. Wang, Y. Wang, C. Wang, Y. Zeng, J. Li, S. Zhang, W. Huang, Adaptable invisibility management using kirigami-inspired transformable metamaterials, Research (2021).
[19] H. Bai, S. Li, R.F. Shepherd, Elastomeric haptic devices for virtual and augmented reality, Adv. Funct. Mater. 31 (39) (2021) 2009364.
[20] O. Bau, U. Petrevski, W. Mackay, BubbleWrap: a textile-based electromagnetic haptic display, in: CHI'09 Extended Abstracts on Human Factors in Computing Systems, Association for Computing Machinery, 2009, pp. 3607–3612.
[21] W. Kang, M. Xu, W. Yao, J. Zhang, Lock-in mechanism of flow over a low-Reynolds-number airfoil with morphing surface, Aerosp. Sci. Technol. 97 (2020) 105647.
[22] E. Thompson, A. Goza, Surface morphing for aerodynamic flows at low and stalled angles of attack, Phys. Rev. Fluids 7 (2) (2022) 024703.
[23] V.J. Shinde, D.V. Gaitonde, J.J. McNamara, Supersonic turbulent boundary-layer separation control using a morphing surface, AIAA J. 59 (3) (2021) 912–926.
[24] D. Foresti, G. Sambatakakis, S. Bottan, D. Poulikakos, Morphing surfaces enable acoustophoretic contactless transport of ultrahigh-density matter in air, Sci. Rep. 3 (1) (2013) 3176.
[25] K. Liu, F. Hacker, C. Daraio, Robotic surfaces with reversible, spatiotemporal control for shape morphing and object manipulation, Sci. Robotics 6 (53) (2021) eabf5116.
[26] J. Wang, M. Sotzing, M. Lee, A. Chortos, Passively addressed robotic morphing surface (PARMS) based on machine learning, Sci. Adv. 9 (29) (2023) eadg8019.
[27] Y. Park, G. Vella, K.J. Loh, Bio-inspired active skins for surface morphing, Sci. Rep. 9 (1) (2019) 18609.
[28] S. An, Y. Cao, H. Jiang, A mechanically robust and facile shape morphing using tensile-induced buckling, Sci. Adv. 10 (21) (2024) eado8431.
[29] T. Gao, J. Bico, B. Roman, Pneumatic cells toward absolute Gaussian morphing, Science 381 (6660) (2023) 862–867.
[30] E. Siéfert, E. Reyssat, J. Bico, B. Roman, Bio-inspired pneumatic shape-morphing elastomers, Nat. Mater. 18 (1) (2019) 24–28.
[31] S. Hu, X. Cao, T. Reddyhoff, X. Ding, X. Shi, D. Dini, A.J. deMello, Z. Peng, Z. Wang, Pneumatic programmable superrepellent surfaces, Droplet 1 (1) (2022) 48–55.
[32] M.K. Blees, A.W. Barnard, P.A. Rose, S.P. Roberts, K.L. McGill, P.Y. Huang, A.R. Ruyack, J.W. Kevek, B. Kobrin, D.A. Muller, et al., Graphene kirigami, Nature 524 (7564) (2015) 204–207.
[33] P. Shi, Y. Chen, J. Feng, P. Sareh, Highly stretchable graphene kirigami with tunable mechanical properties, Phys. Rev. E 109 (3) (2024) 035002.
[34] A. Moura, B. Ipaves, D.S. Galvao, P.A. da Silva Autreto, Ballistic properties of highly stretchable graphene Kirigami pyramid, Comput. Mater. Sci. 232 (2024) 112558.
[35] X. Yang, C. Forró, T.L. Li, Y. Miura, T.J. Zaluska, C.-T. Tsai, S. Kanton, J.P. McQueen, X. Chen, V. Mollo, et al., Kirigami electronics for long-term electrophysiological recording of human neural organoids and assembloids, Nature Biotechnol. (2024) 1–8.
[36] B.M. Li, I. Kim, Y. Zhou, A.C. Mills, T.J. Flewwellin, J.S. Jur, Kirigami-inspired textile electronics: KITE, Adv. Mater. Technol. 4 (11) (2019) 1900511.
[37] F. Zhuo, J. Zhou, Y. Liu, J. Xie, H. Chen, X. Wang, J. Luo, Y. Fu, A. Elmarakbi, H. Duan, Kirigami-inspired 3D-printable mxene organohydrogels for soft electronics, Adv. Funct. Mater. 33 (52) (2023) 2308487.
[38] R. Wei, H. Li, Z. Chen, Q. Hua, G. Shen, K. Jiang, Revolutionizing wearable technology: advanced fabrication techniques for body-conformable electronics, Npj Flex. Electron. 8 (1) (2024) 1–20.
[39] R. Zhao, S. Lin, H. Yuk, X. Zhao, Kirigami enhances film adhesion, Soft Matter 14 (13) (2018) 2515–2525.
[40] D. Hwang, C. Lee, X. Yang, J.M. Pérez-González, J. Finnegan, B. Lee, E.J. Markvicka, R. Long, M.D. Bartlett, Metamaterial adhesives for programmable adhesion through reverse crack propagation, Nat. Mater. 22 (8) (2023) 1030–1038.
[41] D.-G. Hwang, K. Trent, M.D. Bartlett, Kirigami-inspired structures for smart adhesion, ACS Appl. Mater. Interfaces 10 (7) (2018) 6747–6754.
[42] Y. Morikawa, S. Yamagiwa, H. Sawahata, R. Numano, K. Koida, M. Ishida, T. Kawano, Ultrastretchable Kirigami bioprobes, Adv. Healthc Mater. 7 (3) (2018) 1701100.
[43] Z. Liu, H. Du, J. Li, L. Lu, Z.-Y. Li, N.X. Fang, Nano-Kirigami with giant optical chirality, Sci. Adv. 4 (7) (2018) eaat4436.
[44] S. Chen, J. Chen, X. Zhang, Z.-Y. Li, J. Li, Kirigami/origami: unfolding the new regime of advanced 3D microfabrication/nanofabrication with "folding", Light.: Sci. Appl. 9 (1) (2020) 75.
[45] H. Zhang, J. Paik, Kirigami design and modeling for strong, lightweight metamaterials, Adv. Funct. Mater. 32 (21) (2022) 2107401.
[46] Y. Sun, W. Ye, Y. Chen, W. Fan, J. Feng, P. Sareh, Geometric design classification of kirigami-inspired metastructures and metamaterials, in: Structures, Vol. 33, Elsevier, 2021, pp. 3633–3643.
[47] J. Tao, H. Khosravi, V. Deshpande, S. Li, Engineering by cuts: how Kirigami principle enables unique mechanical properties and functionalities, Adv. Sci. 10 (1) (2023) 2204733.
[48] M. Isobe, K. Okumura, Initial rigid response and softening transition of highly stretchable kirigami sheet materials, Sci. Rep. 6 (1) (2016) 24758.

[49] G. Chaudhary, L. Niu, M. Lewicka, Q. Han, L. Mahadevan, Geometric mechanics of random kirigami, 2021, arXiv preprint arXiv:2112.13699.
[50] B. Wu, W. Jiang, J. Jiang, Z. Zhao, Y. Tang, W. Zhou, W. Chen, Wave manipulation in intelligent metamaterials: recent progress and prospects, Adv. Funct. Mater. 34 (29) (2024) 2316745.
[51] I.M. Imani, H.S. Kim, J. Shin, D.-G. Lee, J. Park, A. Vaidya, C. Kim, J.M. Baik, Y.S. Zhang, H. Kang, et al., Advanced ultrasound energy transfer technologies using metamaterial structures, Adv. Sci. 11 (31) (2024) 2401494.
[52] A.P. Paul, K. Nayak, L.C. Sydnor, N. Kalantaryardebily, K.M. Parcetich, D.G. Miner, Q.E. Wafford, J.E. Sullivan, N. Gurari, A scoping review on examination approaches for identifying tactile deficits at the upper extremity in individuals with stroke, J. NeuroEng. Rehabil. 21 (1) (2024) 99.
[53] C. Bermejo, P. Hui, A survey on haptic technologies for mobile augmented reality, ACM Comput. Surv. 54 (9) (2022) 1–35.
[54] R.V. Patel, S.F. Atashzar, M. Tavakoli, Haptic feedback and force-based teleoperation in surgical robotics, Proc. IEEE 110 (7) (2022) 1012–1027.
[55] I. El Rassi, J.-M. El Rassi, A review of haptic feedback in tele-operated robotic surgery, J. Med. Eng. Technol. 44 (5) (2020) 247–254.
[56] C. Kimball, L.-W. Tsai, Modeling of flexural beams subjected to arbitrary end loads, J. Mech. Des. 124 (2) (2002) 223–235.
[57] T. Beléndez, C. Neipp, A. Beléndez, Largeand small deflections of a cantilever beam, Eur. J. Phys. 23 (3) (2002) 371.
[58] P. Khatak, et al., Laser cutting technique: A literature review, Mater. Today: Proc. 56 (2022) 2484–2489.
[59] D.A. Bartholomeusz, R.W. Boutté, J.D. Andrade, Xurography: rapid prototyping of microstructures using a cutting plotter, J. Microelectromech. Syst. 14 (6) (2005) 1364–1374.
[60] K. Melde, A.G. Mark, T. Qiu, P. Fischer, Holograms for acoustics, Nature 537 (7621) (2016) 518–522.
[61] R.K. Mueller, Acoustic holography, Proc. IEEE 59 (9) (1971) 1319–1335.
[62] W. Veronesi, J.D. Maynard, Nearfield acoustic holography (NAH) II. Holographic reconstruction algorithms and computer implementation, J. Acoust. Soc. Am. 81 (5) (1987) 1307–1322.
[63] B. Hildebrand, An Introduction to Acoustical Holography, Springer Science & Business Media, 2012.
[64] A. Sallam, Multi-Functional Holographic Acoustic Lenses for Modulating Low-To High-Intensity Focused Ultrasound (Dissertation), Virginia Tech, 2024.
[65] M.D. Brown, Phase and amplitude modulation with acoustic holograms, Appl. Phys. Lett. 115 (5) (2019).
[66] S. Castiñeira-Ibáñez, D. Tarrazó-Serrano, A. Uris, C. Rubio, O.V. Minin, I.V. Minin, Cylindrical 3D printed configurable ultrasonic lens for subwavelength focusing enhancement, Sci. Rep. 10 (1) (2020) 20279.
[67] G. Maimbourg, A. Houdouin, T. Deffieux, M. Tanter, J.-F. Aubry, 3D-printed adaptive acoustic lens as a disruptive technology for transcranial ultrasound therapy using single-element transducers, Phys. Med. Biol. 63 (2) (2018) 025026.
[68] A. Sallam, C. Cengiz, M. Pewekar, E. Hoffmann, W. Legon, E. Vlaisavljevich, S. Shahab, Gradient descent optimization of acoustic holograms for transcranial focused ultrasound, J. Appl. Phys. 136 (14) (2024).
[69] V.F.-G. Tseng, S.S. Bedair, N. Lazarus, Phased array focusing for acoustic wireless power transfer, IEEE Trans. Ultrason. Ferroelectr. Freq. Control 65 (1) (2017) 39–49.
[70] Y. Xie, C. Shen, W. Wang, J. Li, D. Suo, B.-I. Popa, Y. Jing, S.A. Cummer, Acoustic holographic rendering with two-dimensional metamaterial-based passive phased array, Sci. Rep. 6 (1) (2016) 35437.
[71] J. Zhao, H. Ye, K. Huang, Z.N. Chen, B. Li, C.-W. Qiu, Manipulation of acoustic focusing with an active and configurable planar metasurface transducer, Sci. Rep. 4 (1) (2014) 6257.
[72] K.K. Shung, High frequency ultrasonic imaging, J. Med. Ultrasound 17 (1) (2009) 25–30.
[73] G. Lockwood, D. Turnball, D. Christopher, F.S. Foster, Beyond 30 MHz [applications of high-frequency ultrasound imaging], IEEE Eng. Med. Biol. Mag. 15 (6) (1996) 60–71.
[74] R.H. Silverman, Focused ultrasound in ophthalmology, Clin. Ophthalmol. (2016) 1865–1875.
[75] G. Darmani, T. Bergmann, K.B. Pauly, C. Caskey, L. De Lecea, A. Fomenko, E. Fouragnan, W. Legon, K. Murphy, T. Nandi, et al., Non-invasive transcranial ultrasound stimulation for neuromodulation, Clin. Neurophysiol. 135 (2022) 51–73.
[76] S.D. Mellin, G.P. Nordin, Limits of scalar diffraction theory and an iterative angular spectrum algorithm for finite aperture diffractive optical element design, Opt. Express 8 (13) (2001) 705–722.
[77] C.-Y. Shen, Y. Cheng, S.-H. Huang, Y.-P. Huang, P-90: Image enhancement of 3D holographic head-up display using multi-constraints angular spectrum algorithm, in: SID Symposium Digest of Technical Papers, Vol. 50, Wiley Online Library, 2019, pp. 1576–1579, No. 2.
[78] M. Xu, J. Wang, W.S. Harley, P.V. Lee, D.J. Collins, Programmable acoustic holography using medium-sound-speed modulation, Adv. Sci. 10 (23) (2023) 2301489.
[79] W.C. Young, R.G. Budynas, A.M. Sadegh, et al., Roark's Formulas for Stress and Strain, vol. 7, McGraw-hill, New York, 2002.
[80] F.H. Giraud, S. Joshi, J. Paik, Haptigami: A fingertip haptic interface with vibrotactile and 3-DoF cutaneous force feedback, IEEE Trans. Haptics 15 (1) (2022) 131–141.
[81] C.-H. Yeh, F.-C. Su, Y.-S. Shan, M. Dosaev, Y. Selyutskiy, I. Goryacheva, M.-S. Ju, Application of piezoelectric actuator to simplified haptic feedback system, Sens. Actuators A: Phys. 303 (2020).
[82] N. Gurari, G. Baud-Bovy, Customization, control, and characterization of a commercial haptic device for high-fidelity rendering of weak forces, J. Neurosci. Methods 235 (2014) 169–180.
[83] K. Iiyoshi, J.H. Lee, A.S. Dalaq, S. Khazaaleh, M.F. Daqaq, G. Korres, M. Eid, Origami-inspired haptics: A literature review, IEEE Access 12 (2024) 33309–33327.
[84] N. Kalantaryardebily*, A.C. Feldbush*, R. Faubion-Trejo, J. Lisinski, N.A. Reddy, M.G. Bright, S.M. LaConte, N. Gurari, Development and testing of an MR-compatible tactile stimulator system: Application for individuals with a brain injury, J. Neurosci. Methods 424 (2025) 110583.
[85] R. Bekrater-Bodmann, J. Foell, M. Diers, S. Kamping, M. Rance, P. Kirsch, J. Trojan, X. Fuchs, F. Bach, H.K. Çakmak, H. Maaß, H. Flor, The importance of synchrony and temporal order of visual and tactile input for illusory limb ownership experiences – An fmri study applying virtual reality, PLoS One 9 (1) (2014) e87013.
[86] M. Brickwedde, M.D. Schmidt, M.C. Krüger, H.R. Dinse, 20 Hz steady-state response in somatosensory cortex during induction of tactile perceptual learning throug LTP-like sensory stimulation, Front. Hum. Neurosci. 14 (2020) 257.
[87] N. Kalantaryardebily, A.C. Feldbush, A. Kahak, S. Li, N. Gurari, Novel compact tactile stimulator with sensing: Designed for individuals with a brain injury and MRI, IEEE Trans. Med. Robotics Bionics (2025) in revision.
[88] A.C. Feldbush, N. Kalantaryardebily, N.A. Reddy, R. Faubion-Trejo, J. Soldate, J. Lisinski, M.G. Bright, N. Gurari, S.M. LaConte, Tactile pressure evokes a biphasic BOLD response in ipsilateral primary somatosensory cortex, 2025, Published in bioRxiv [Preprint].
[89] K.J. Bretz, A. Jobbagy, K. Bretz, Force measurement of hand and fingers, Biomech. Hung. (2010).
[90] R. Gassert, A. Yamamoto, D. Chapuis, L. Dovat, H. Bleuler, E. Burdet, Actuation methods for applications in MR environments, Concepts Magn. Reson. Part B: Magn. Reson. Eng. 29B (4) (2006) 191–209.